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Discrete Optimization

A cutting plane algorithm for the unrelated parallel machine scheduling problem

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Abstract

This paper deals with the polyhedral structure of the scheduling problem $R//C_{\max}$. Strong valid inequalities are identified for fixed values of the maximum completion time and are used to build a cutting plane scheme from which an exact algorithm and an approximation algorithm are developed. The main algorithm includes a preprocessing phase to compute an upper bound with the list scheduling algorithm of Davis and Jaffe and a lower bound from the preemptive version of the problem. Computational results show that the proposed exact algorithm gives an optimal solution for almost all tested cases, within the fixed time and memory limits. For the few cases where the limit has been exceeded, rather good solutions are obtained. Computational requirements of both algorithms are reported for a collection of test problems and the quality of the schedules provided by the approximation algorithm is measured. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

In the scheduling problem $R//C_{\max}$, n independent and non-preemptive jobs must be processed on m different machines. A job may be processed on each of the m machines but the processing time depends on the machine on which the job is performed. Each of the jobs has to be processed on one machine. The problem is to find a schedule whose makespan is minimum.

It is common to find real problems that could be modeled by an instance of $R//C_{\max}$. For example, on production lines, there is usually more than one machine that can carry out the production tasks. Other examples are docking systems for ships, share-time systems, multiprocessor computers, etc. Many other problems could be mentioned when speaking about scarce resources or machines dedicated to the production of some goods, or carrying out tasks.

Due to the NP-hardness of the problem at hand [5], approximation algorithms have been intensively developed. Exact algorithms have been

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proposed by van de Velde [29] and Martello et al. [17]. The former is based on surrogate relaxation and duality, and the latter on Lagrangian relaxations and additive techniques.

The aim of this paper is to present a cutting plane scheme built from the identification of a new valid inequality that applies to the subset of the solutions constrained by a maximum value of the makespan. That valid inequality leads to an exact algorithm and to an approximation algorithm that use linear programming (LP) formulations with binary decision variables. Note that a similar technique was applied successfully to the scheduling problem $P//C_{\max}$ [19] (where P means that the machines are identical). We have given polyhedral approaches a trial-run because Polyhedral Combinatorics have turned out to be practically successful computational tools in many hard Combinatorial Optimization problems like the classical Traveling Salesman Problem [22].

Combining heuristics and cutting plane techniques, we analyze the structure of the corresponding polytope. Valid inequalities are identified for a fixed value of $C_{\max}$. Starting with an initial lower bound on the makespan, its value is increased by one unit at each iteration until either an integer solution is found, no more valid inequalities may be added or the upper bound value is reached. The core of the exact algorithm and the approximation algorithm is a cutting plane method that uses the simplex method to solve the LP relaxations. Computational results show that almost all tested instances could be optimally solved by the exact algorithm. Moreover, for all tested instances the approximation algorithm has provided near optimal solutions. When the exact algorithm exceeds the fixed time and memory limits, the best solution already obtained is selected, which seems to be an interesting feature of the work presented in this paper since some instances have remained unsolved by the exact algorithms published up to now [17,29].

In this first section, the scheduling problem and previous results are looked at. The cutting plane scheme is presented in Section 2. A valid inequality for the solutions constrained by a maximum makespan value is proposed in Section 3. Section 4 gives an exact and an approximation algorithm. Computational experiments are shown and discussed in Section 5. We conclude with some suggestions for further research.

1.1. Problem definition and previous results

Consider a set J of n independent and non-preemptable jobs J_i to be processed on m unrelated machines $M_1, \ldots, M_m$. Each job may be processed on any of the m machines but the processing time p_{ij} of job J_i depends on the machine M_j that executes that job. The objective is to find an assignment of the n jobs to the machines that minimizes the makespan $C_{\max}$.

Let x_{ij} be the $0/1$ assignment variable, which is equal to 1 (respectively, 0) if job J_i is assigned (respectively, not assigned) to machine M_j, a Mixed Integer Programming (MIP) formulation of the problem is as follows:

$$\min y$$

$$\sum_{j=1}^{m} x_{ij} = 1 \quad i = 1, \ldots, n, \tag{1}$$

$$y - \sum_{i=1}^{n} p_{ij} x_{ij} \geq 0 \quad j = 1, \ldots, m. \tag{2}$$

The constraint in (1) associated with job J_i simply means that J_i must be processed by one machine. The constraint in (2) associated with machine M_j means that the makespan is at least the value of the load of that machine. This formulation has $m + n$ constraints and $mn + 1$ variables. Since for given values x_{ij}^0 of the assignment variables, the minimum value of y is $\max_{j=1,\ldots,m} \sum_{i=1}^{n} p_{ij} x_{ij}^0$, the variable y does not have to be restricted to being an integer.

The decision variant of $R//C_{\max}$ is known to be NP-complete in the strong sense since 3-PARTITION is its special case. Moreover Lenstra and Shmoys [15] proved that unless $P = NP$, no polynomial approximation algorithms with a performance ratio of less than 4/3 could exist for $R//C_{\max}$.

Only a few scheduling problems with identical parallel machines have been shown to be polynomial. Two such well-known problems are $P/pmtn/C_{\max}$, where tasks may be preempted, and $P//\sum_j C_j$, where the sum of the tasks completion

times has to be minimized. For a full review, see Cheng and Sin [2], Lawler et al. [14] or Mokotoff [18]. For hard problems like the problem at hand, research has focused on securing optimal solutions by enumerative methods or on developing approximation algorithms.

Most scheduling problems with non-identical machines are also NP-hard. Among the numerous papers dealing with such problems, we may cite the following references that concern the minimization of the makespan. Exact and approximate algorithms are presented in [21]. Branch and Bound (B&B) procedures are developed by Stern [28] and Martello et al. [17]. Horowitz and Sahni [9] propose general Dynamic Programming schemes. Van de Velde [29] presents an exact algorithm and an approximation algorithm based on surrogate relaxation and duality.

Let us now mention relevant results referring to approximation algorithms. Ibarra and Kim [10] present a comparative study of five heuristic algorithms for scheduling independent jobs and conclude that, as a rule, the worst-case performance ratio is bad. Friesen and Langston [4] apply the results of MultiFit [1] to uniform machines. In [8,16,24], two-stage approximation algorithms, which have been shown to be highly efficient, have been developed. In the first stage, LP is used to obtain a fractional approximation of the solution while in the second stage the fractional schedule is transformed into a feasible integer schedule by